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Low but Increasing Prevalence of Reduced Beta-lactam Susceptibility Among Invasive Group B Streptococcal Isolates, US Population-Based Surveillance, 1998–2018

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Background. Invasive group B *Streptococcus* (iGBS) isolates with mutations in the *pbp2x* gene that encodes penicillin binding protein 2x can have reduced beta-lactam susceptibility (RBLS) when susceptible by Clinical and Laboratory Standards Institute (CLSI) criteria. We assessed the emergence and characteristics of RBLS strains in US iGBS isolates.

Methods. We analyzed iGBS isolates from 8 multistate population-based surveillance sites from 1998 to 2018. During 1998–2014, phenotypic antimicrobial susceptibility was determined by broth microdilution; criteria for 6 antibiotics were used to identify RBLS, followed by whole-genome sequencing (WGS). WGS for all isolates was added in 2015; we used phenotypic and genotypic results of >2000 isolates to validate phenotypic RBLS criteria and genotypic predictions. Since 2016, WGS has been used to screen for RBLS with broth microdilution confirmation of predicted RBLS isolates.

Results. Of 28 269 iGBS isolates, 28 (0.1%) were nonsusceptible by CLSI criteria; 137 (0.5%) met RBLS criteria. RBLS isolates were detected in all Active Bacterial Core surveillance sites. The RBLS proportion increased, especially since 2013 (odds ratio, 1.17; 95% CI, 1.03–1.32); the proportion that were nonsusceptible remained stable.

Conclusions. The RBSL proportion was low but increasing among US iGBS isolates. Ongoing monitoring is needed to detect emerging threats to prevention and treatment of GBS infections.

Keywords. group B *Streptococcus*; penicillin binding protein 2x; reduced beta-lactam susceptibility; whole-genome sequencing.

In the United States, invasive group B *Streptococcus* (GBS) infection remains an important cause of illness and death among young infants, with an overall case fatality rate of 5.4%–6.9% [1], and a cause of disease that rivals *Streptococcus pneumoniae* in national case burden among nonpregnant older adults and persons with underlying medical conditions [2]. Beta-lactam antibiotics such as penicillin or ampicillin are recommended for intrapartum antibiotic prophylaxis (IAP) [3, 4] and treatment of invasive GBS infection [4, 5]. GBS isolates with reduced susceptibility to beta-lactam antibiotics, first reported in the 1990s [5], have been linked to point mutations in the *pbp2x* gene that encode penicillin binding protein 2x (PBP2x) [6, 7]. In *S. pneumoniae,* modification of the *pbp2x* gene is required for resistance to penicillin. Based on the susceptibility criteria defined by the Clinical and Laboratory Standards Institute (CLSI), penicillin-nonsusceptible invasive GBS isolates are rare in the United States [8]; however, *pbp2x* point mutations associated with reduced susceptibility to beta-lactam antibiotics may be present even when isolates remain clinically susceptible to penicillin [6, 9, 10]. Thus, detection of GBS isolates with first-step *pbp2x* mutations through routine surveillance may allow for early public health intervention before clinical resistance emerges.

In 2015, Active Bacterial Core surveillance (ABCs), which conducts population-based surveillance for invasive GBS disease in the United States, introduced whole-genome sequencing (WGS) for GBS isolates. All ABCs GBS isolates collected in 2015 underwent both WGS and broth microdilution susceptibility testing [9]. This allowed minimum inhibitory concentration (MIC) thresholds for 6 beta-lactam antibiotics, different from those defined by the CLSI for clinical susceptibility [11], to be validated for their ability to detect first-step *pbp2x* mutations conferring reduced beta-lactam susceptibility (RBLS) [6]. Use of these validated MIC breakpoints and WGS allowed us to analyze 20 years of ABCs GBS surveillance to describe trends in the proportion of isolates with RBLS, phenotypic and genetic characteristics of RBLS strains, and characteristics of patients with invasive GBS infection caused by RBLS strains.

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METHODS

Surveillance

ABCs conducts active, population- and laboratory-based surveillance for invasive GBS disease in select counties in 10 states across the United States, and isolates are sent to the Centers for Disease Control and Prevention (CDC) from a subset of sites. ABCs surveillance methods have been described in detail elsewhere [12, 13]. In 1998, GBS isolates were sent from Georgia (20 counties), New York (7 counties), Oregon (3 counties), and Minnesota (all counties) to the CDC. By 2018, sites submitting GBS isolates expanded to include California (3 counties), Colorado (5 counties), Georgia (20 counties), New York (15 counties, only from children aged <90 days), Oregon (3 counties), and statewide for Maryland, Minnesota, and New Mexico [14]. This analysis was restricted to GBS cases from the 8 sites with isolates available for evaluation. As information on antibiotic administration is not routinely collected, ABCs sites performed additional medical record abstraction once cases of RBLS infection were identified.

Definitions

An invasive case was defined as isolation of GBS from a normally sterile site (eg, blood, cerebrospinal fluid) from a patient residing in the ABCs catchment area. The index GBS case refers to the invasive GBS case captured by ABCs. RBLS, a sensitive surveillance definition to capture isolates with modified *pbp2x* precursors to clinical nonsusceptibility, was defined as minimum inhibitory concentration (MIC) values that exceeded the predefined breakpoints for at least 1 of 6 beta-lactam antibiotics (ampicillin MIC ≥ 0.5 μg/mL, penicillin MIC ≥ 0.25 μg/mL, cefotaxime MIC ≥ 0.25 μg/mL, ceftizoxime MIC ≥ 1 μg/mL, cefazolin MIC ≥ 1 μg/mL, or cefoxitin MIC ≥ 16 μg/mL) in a GBS isolate [9]. Beta-lactam nonsusceptibility (NS) was defined as an MIC exceeding CLSI breakpoints for beta-lactam susceptibility in a GBS isolate: ampicillin MIC > 0.25 μg/mL, penicillin MIC > 0.12 μg/mL, or cefotaxime MIC > 0.5 μg/mL [15]. PBP2x types were determined based on the amino acid sequence difference from the major allele, PBP2x type 1 (Supplementary Table 1) [9].

Specimen Collection and Testing

GBS isolates from ABCs sites, except for Minnesota, were serotyped and subjected to antimicrobial susceptibility testing at the CDC *Streptococcus* laboratory in Atlanta, Georgia; all isolates from Minnesota were tested at the Minnesota Public Health Laboratory. Through 2014, MICs were determined by broth microdilution only. In 2015, isolates (n = 1975 from 2015 and n = 56 RBLS isolates from 2003–2014) underwent both broth microdilution and WGS (Supplementary Data) [9]. Of the 2004 isolates that did not exceed the RBLS criteria, 97% were PBP2x types 1–5, and none of the isolates with PBP2x types 1–5 (n = 1949) exceeded the RBLS criteria [9]. Therefore, after 2015, all isolates underwent WGS, but only those with PBP2x transpeptidase amino acid sequence types 6 and above (PBP2x types ≥6) or those whose PBP2x types could not be identified by WGS (<1% of isolates) underwent broth microdilution. Broth microdilution results continued to be available for all isolates from Minnesota (>20% of all GBS isolates submitted). Broth microdilution was repeated on all RBLS isolates from 1998 to 2018 using a standardized panel. PBP2x types, multilocus sequence typing, and capsular serotypes were determined by WGS based on previously described methods [8, 9]. Isolates from 1998–2014 were serotyped by latex agglutination using rabbit antisera to GBS capsular polysaccharide types Ia, Ib, and II–IX [16]. During 2007–2014, polymerase chain reaction (PCR) serotyping was performed on isolates that were nontypeable by latex agglutination [17]. Isolates were considered nontypeable if PCR did not identify the serotype. The established bioinformatics pipeline [9] identified *pbp2x* point mutants and sequence types (STs) that are in the global GBS database (https://pubmlst.org/sagalactiae/). Phylogenetic analysis was performed as described in the Supplementary Data.

Statistical Analysis

We performed analyses of GBS cases and associated isolates and tested the associations of RBSL or NS cases with a list of explanatory variables including demographic, behavioral, and clinical factors. We computed medians and interquartile ranges (IQRs) for the continuous explanatory variables. We conducted the Fisher exact test for categorical variables with 2 categories or Monte Carlo Simulation for the Fisher exact test for variables with >2 categories (2-sided $P < .05$ was considered statistically significant). Descriptive analysis of adult cases focused on adults aged ≥30 years, as there were no RBLS or NS cases among younger adults. Simple logistic regression was used to assess the change in the proportion of NS or RBLS isolates over time. Multivariable logistic regression adjusted for state and age to account for changes in sites, and age groups eligible for specimen shipment over time were used to assess trends over time [14]. All analyses were performed using SAS, version 9.4 (SAS Institute Inc.).

Ethics

The project was reviewed in accordance with CDC human research protection procedures and was determined to be nonresearch. ABCs sites reviewed the protocol and obtained local institutional review board approval where required.

RESULTS

Overall Trends and Characteristics of Invasive GBS Cases

A total of 28 269 of 33 337 (84.8%) GBS cases reported in surveillance catchment areas submitting isolates had susceptibility

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data on isolates (either from broth microdilution or WGS, as described in the "Methods") and were included in this analysis. Information on specimen source is included in the Supplementary Data. Most (20 475/28 269; 72.4%) invasive GBS cases were reported in 3 ABCs sites, Georgia, Maryland, and Minnesota, in adults aged ≥45 years (20 919/28 267; 74.0%) (Table 1). In infants aged <90 days, primary bacteremia (bacteremia) was the most common presenting syndrome (72.6%). Bacteremia was also the most common syndrome in adults aged ≥30 years, but was less frequent compared with infants (37.8%). Nearly half (48.6%) of cases in adults occurred in people with diabetes. The proportion of cases identified in older age groups (especially adults aged ≥45 years) and adults with underlying chronic conditions such as diabetes and chronic lung disease increased over time (Supplementary Table 2). In both young infants and adults aged ≥30 years, the proportion of cases with bacteremia decreased over time. In contrast, the proportion with meningitis in infants, and the proportion with cellulitis and osteomyelitis in adults, increased over time.

Of the 28 269 isolates, 137 (0.5%) met the RBLS criteria, and 28 (0.1%) met the NS criteria (Figure 1). The proportion of RBLS isolates increased significantly from 1998 to 2018 (odds ratio [OR], 1.10; 95% CI, 1.06–1.14) especially since 2013 (OR, 1.17; 95% CI, 1.03–1.32). Since most isolates from before 2014 were missing susceptibility data for ceftizoxime, we also assessed the change in the proportion of RBLS isolates without ceftizoxime. The overall trend did not change (OR, 1.09; 95% CI, 1.05–1.13) (Supplementary Figure 1). The overall trend also did not change after excluding California, which started submitting isolates from 2014 (data not shown). There was no significant increase in the proportion of NS isolates by year (OR, 1.03; 95% CI, 0.96–1.11).

Description of RBLS Cases

RBLS isolates were detected in all 8 sites, and NS isolates were detected in all but 1 (Oregon) site. Among the 8 sites, California, which started submitting specimens in 2014, had the highest proportion of RBLS isolates (1.2% overall), but not if limited to 2014–2018 (Supplementary Table 3). NS and RBLS cases were identified in infants aged <90 days (NS: 2 cases; RBLS: 11 cases) and in adults aged ≥30 years (NS: 26 cases; RBLS: 126 cases) (Table 1). The number of RBLS cases in adults aged ≥30 years increased after 2014 (Supplementary Table 4). The age distribution among RBLS cases was not statistically significantly different before and after 2014. There were 4 deaths identified among RBLS cases (2.9%) and none among NS cases (Table 1). All 4 deaths were in adults after 2014, and all but 1 were in adults aged ≥65 years.

Of the 2 NS cases among infants aged <90 days, 1 was in an infant aged <7 days with bacteremia whose mother received IAP; the other was in an infant aged 7–89 days with osteomyelitis whose mother did not receive IAP. Of the 11 RBLS cases, 5 were in infants aged <7 days, including 1 whose mother received IAP; 6 were in infants aged 7–89 days, including 3 whose mothers received IAP. Meningitis (54.6%) and bacteremia (27.3%) were common types of infection among RBLS cases. The proportion of RBLS cases among those with bacteremia was much lower (0.12%) compared with those with meningitis (0.95%) or osteomyelitis (3.70%) (Table 1).

In adults, those with bacteremia had the lowest proportion of both NS (0.02%) and RBLS (0.39%) cases, and those with osteomyelitis had the highest proportion of NS (0.37%) and RBLS (1.00%) cases. By underlying conditions, adults with diabetes had higher proportions of both NS (0.18% vs 0.04%; $P = .001$) and RBLS (0.64% vs 0.44%; $P = .03$) infection compared with those without diabetes. Adults who are current smokers (0.93% vs 0.47%; $P = .003$) and those with chronic skin breakdown (0.97% vs 0.55%; $P = .02$) had a higher proportion of RBLS infection, but there were no statistically significant differences in the proportion of NS isolates (Table 1).

Antibiotic Treatment Course and History of Prior Exposure

Information on antibiotic use was available for 21/28 (75.0%) NS and 115/137 (83.9%) RBLS cases (Table 2). All NS cases and 113/115 (98.3%) RBLS cases received antibiotic treatment for the index GBS case. Of those who received treatment, the median duration of antibiotic treatment (IQR) was 15 (10–30) days for NS cases and 14 (9–18) days for RBLS cases, excluding 17 with unknown duration of treatment or those who died. In adults aged ≥30 years, 8/19 (42.1%) NS cases and 37/106 (34.9%) RBLS cases had known beta-lactam antibiotic treatment history in the year before the index GBS case; the median number of days of prior antibiotic exposure was longer in NS cases (median [IQR], 25.5 [8–33] days) compared with RBLS cases (median [IQR], 12.5 [2–28.5] days). In adults with RBLS infection, there was a longer duration of beta-lactam antibiotics in the preceding year in those with osteomyelitis (median [IQR], 21 [18–32] days) compared with those with bacteremia (median [IQR], 5.5 [1–15] days).

Characterization of RBLS Isolates

Most RBLS isolates (83/137; 60.6%) had reduced susceptibility to cefoxitin or ceftizoxime, or both (Table 3). Among isolates meeting both NS and RBLS criteria, 2 isolates had reduced susceptibility to ampicillin only, and the rest had RBLS to ≥3 individual beta-lactam antibiotics. There were no isolates that met NS criteria for cefotaxime (>0.5 µg/mL), although several RBLS isolates had cefotaxime MIC values ranging from 0.25 µg/mL to 0.5 µg/mL. Of the 135 RBLS isolates with PBP2x type information available, 15 (11.1%) were PBP2x types 1–5, which due to their high frequency and rare association with isolates with RBLS, are PBP2x types considered not to have *pbp2x* mutations associated with RBLS; 8/15 (53.3%) met the RBLS criteria based on cefoxitin, ceftizoxime, or cefotaxime only, and all but 2 had MIC values that were at the breakpoint

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Table 1. Characteristics of Invasive GBS Cases That Had Isolates Tested for Beta-lactam Susceptibility, ABCs 1998–2018

Characteristics	Total Number of Cases With Isolate Information (%) (n = 28 269)	Proportion of Isolates That Were NS, by Row Characteristic, %	NS Isolates (%) (n = 28)	Beta-lactam Susceptible Isolates (%) (n = 28 241)	*P* Value	Proportion of Isolates That Were RBLS, by Row Characteristic, %	RBLS Isolates (%) (n = 137)	Non-RBLS Isolates (%) (n = 28 132)	*P* Value
State					**.04**				.11
California[a]	955 (3.4)	0.21	2 (7.1)	953 (3.4)		1.15	11 (8.0)	944 (3.4)	
Colorado[b]	1887 (6.7)	0.11	2 (7.1)	1888 (6.7)		0.48	9 (6.6)	1878 (6.7)	
Georgia[c]	5246 (18.6)	0.06	3 (10.7)	5243 (18.6)		0.44	23 (17.0)	5223 (18.6)	
Maryland[d]	7472 (26.4)	0.07	5 (17.9)	7467 (26.4)		0.36	27 (19.7)	7445 (26.5)	
Minnesota[c]	7757 (27.4)	0.10	8 (28.6)	7749 (27.4)		0.54	42 (30.7)	7715 (27.4)	
New Mexico[d]	2063 (7.3)	0.19	4 (14.3)	2059 (7.3)		0.39	8 (5.8)	2055 (7.3)	
New York[e]	992 (3.5)	0.40	4 (14.3)	988 (3.5)		0.60	6 (4.4)	986 (3.5)	
Oregon[c]	1897 (6.7)	0.00	0	1897 (6.7)		0.58	11 (8.0)	1886 (6.7)	
Serotype					.37				.13
IA	6607 (23.7)	0.09	6 (21.4)	6601 (23.7)		0.68	45 (33.1)	6562 (23.7)	
IB	3563 (12.8)	0.03	1 (3.6)	3562 (12.8)		0.39	14 (10.3)	3549 (12.8)	
II	3953 (14.2)	0.08	3 (10.7)	3950 (14.2)		0.30	12 (8.8)	3941 (14.2)	
III	4538 (16.3)	0.13	6 (21.4)	4532 (16.3)		0.51	23 (17.0)	4515 (16.3)	
IV	2330 (8.4)	0.04	1 (3.6)	2329 (8.4)		0.43	10 (7.4)	2320 (8.4)	
V	6174 (22.2)	0.18	11 (39.3)	6163 (22.1)		0.50	31 (22.8)	6145 (22.1)	
Other	711 (2.6)	0.00	0	711 (2.6)		0.14	1 (0.7)	710 (2.6)	
Sex[g]					.25				.44
Male	15 702 (55.6)	0.12	19 (67.9)	15 683 (55.5)		0.52	81 (59.1)	15 621 (55.5)	
Female	12 563 (44.4)	0.07	9 (32.1)	12 554 (44.5)		0.45	56 (40.9)	12 507 (44.5)	
Age group[h]					.79 (for all age groups)				.18 (for all age groups)
<7 d	1710 (6.1)	0.06	1 (3.6)	1709 (6.1)		0.29	5 (3.7)	1705 (6.1)	
7–89 d	1627 (5.8)	0.06	1 (3.6)	1626 (5.8)		0.37	6 (4.4)	1621 (5.8)	
90 d–<1 y	219 (0.8)	0.00	0	219 (0.8)		0.00	0	219 (0.8)	
1–14 y	116 (0.4)	0.00	0	116 (0.4)		0.00	0	116 (0.4)	
15–29 y	976 (3.5)	0.00	0	976 (3.5)		0.00	0	976 (3.5)	
30–44 y	2700 (9.6)	0.15	4 (14.3)	2696 (9.6)	.35 (limited to adults aged ≥30 y)	0.48	13 (9.5)	2687 (9.6)	0.72 (limited to adults aged ≥30 y)
45–64 y	9644 (34.1)	0.13	13 (46.4)	9631 (34.1)		0.50	48 (35.0)	9596 (34.1)	
≥65 y	11 275 (39.9)	0.08	9 (32.1)	11 266 (39.9)		0.58	65 (47.5)	11 210 (39.9)	
Race					.05				.37
White	17 308 (61.2)	0.10	18 (64.3)	17 290 (61.2)		0.53	91 (66.2)	17 217 (61.2)	
Black	6305 (22.3)	0.03	2 (7.2)	6303 (22.3)		0.38	24 (17.5)	6281 (22.3)	
Other	4656 (16.5)	0.17	8 (28.6)	4648 (16.5)		0.47	22 (16.1)	4634 (16.5)	
Recurrent case with the same pathogen?[i]					.70				.15

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Table 1. Continued

Characteristics	Total Number of Cases With Isolate Information (%) (n = 28 269)	Proportion of Isolates That Were NS, by Row Characteristic, %	NS Isolates (%) (n = 28)	Beta-lactam Susceptible Isolates (%) (n = 28 241)	*P* Value	Proportion of Isolates That Were RBLS, by Row Characteristic, %	RBLS Isolates (%) (n = 137)	Non-RBLS Isolates (%) (n = 28 132)	*P* Value
Yes	1764 (6.3)	0.11	2 (7.1)	1762 (6.3)		0.23	4 (3.0)	1760 (6.3)	
No	26 234 (93.6)	0.10	26 (92.9)	26 208 (93.7)		0.50	131 (97.0)	26 103 (93.7)	
Outcome[j]					.26				.09
Survived	26 320 (93.3)	0.11	28 (100)	26 292 (93.3)		0.51	133 (97.1)	26 187 (93.3)	
Died	1895 (6.7)	0	0	1895 (6.7)		0.21	4 (2.9)	1891 (6.7)	
Infants Aged <90 d	Total Number of Isolates With Isolate Information (%) (n = 3337)	Proportion of Isolates That Were NS, by Row Characteristic	NS Isolates (%) (n = 2)	Beta-lactam Susceptible Isolates (%) (n = 3335)	*P* Value	Proportion of Isolates That Were RBLS, by Row Characteristic	RBLS Isolates (%) (n = 11)	Non-RBLS Isolates (%) (n = 3326)	*P* Value
Type of invasive GBS disease[k]					**.02**				**.001**
Bacteremia	2424 (72.6)	0.04	1 (50.0)	2423 (72.6)		0.12	3 (27.3)	2421 (72.8)	
Meningitis	633 (19.0)	0.00	0	633 (19.0)		0.95	6 (54.6)	627 (18.9)	
Osteomyelitis	27 (0.8)	3.70	1 (50.0)	26 (0.8)		3.70	1 (9.1)	26 (0.8)	
Other	253 (7.6)	0.00	0	253 (7.6)		0.40	1 (9.1)	252 (7.6)	
Mother received IAP[l]					>.99				.61
Yes	1275 (43.5)	0.08	1 (50.0)	1274 (43.5)		0.31	4 (36.4)	1271 (43.5)	
No	1657 (56.5)	0.06	1 (50.0)	1656 (56.5)		0.42	7 (63.6)	1650 (56.5)	
Adults Aged ≥30 y	Total Number of Cases With Isolate Information (%) (n = 23 619)	Proportion of Isolates That Were NS, by Row Characteristic	NS Isolates (%) (n = 26)	Beta-lactam Susceptible Isolates (%) (n = 23 593)	*P* Value	Proportion of Isolates That Were RBLS, by Row Characteristic	RBLS Isolates (%) (n = 126)	Non-RBLS Isolates (%) (n = 23 493)	*P* Value
Type of invasive GBS disease[m]					**<.001**				**.008**
Bacteremia	8930 (37.8)	0.02	2 (7.7)	8928 (37.8)		0.39	35 (27.8)	8895 (37.9)	
Cellulitis	5489 (23.2)	0.16	9 (34.6)	5480 (23.2)		0.56	31 (24.6)	5458 (23.2)	
Osteomyelitis	2703 (11.4)	0.37	10 (38.5)	2693 (11.4)		1.00	27 (21.4)	2676 (11.4)	
Pneumonia	2320 (9.8)	0.09	2 (7.7)	2318 (9.8)		0.60	14 (11.1)	2306 (9.8)	
Others	4177 (17.7)	0.07	3 (11.5)	4174 (17.7)		0.45	19 (15.1)	4158 (17.7)	
Underlying condition									
Diabetes mellitus					**.001**				**.03**
Yes	11 480 (48.6)	0.18	21 (80.8)	11 459 (48.6)		0.64	74 (58.3)	11 406 (48.6)	
No	12 139 (51.4)	0.04	5 (19.2)	12 134 (51.4)		0.44	52 (41.3)	12 087 (51.5)	
Chronic heart disease					.83				.43
Yes	7219 (30.6)	0.10	7 (26.9)	7212 (30.6)		0.47	34 (27.0)	7185 (30.6)	
No	16 400 (69.4)	0.12	19 (73.1)	16 381 (69.4)		0.57	92 (72.2)	16 308 (69.4)	
Chronic lung disease					.58				.06
Yes	3548 (15.0)	0.14	5 (19.2)	3543 (15.0)		0.76	27 (21.4)	3521 (15.0)	
No	20 071 (85.0)	0.10	21 (80.8)	20 050 (85.0)		0.50	99 (78.6)	19 972 (85.0)	

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Current smoker[n]					>.99				**.003**
Yes	2808 (12.4)	0.11	3 (12.0)	2805 (12.4)		0.93	26 (22.0)	2782 (12.3)	
No	19 924 (87.6)	0.11	22 (88.0)	19 902 (87.7)		0.47	92 (78.0)	19 832 (87.7)	
Chronic skin breakdown[k]					.29				**.02**
Yes	2372 (14.4)	0.17	4 (23.5)	2368 (14.4)		0.97	23 (23.0)	2349 (14.4)	
No	14 048 (85.6)	0.09	13 (76.5)	14 035 (85.6)		0.55	77 (77.0)	13 971 (85.6)	
Solid organ malignancy[k]					.72				.88
Yes	2238 (13.6)	0.04	1 (5.9)	2237 (13.6)		0.63	14 (14.0)	2224 (13.6)	
No	14 182 (86.4)	0.11	16 (94.1)	14 166 (86.4)		0.61	86 (86.0)	14 096 (86.4)	
Chronic liver disease					.70				.72
Yes	1625 (6.9)	0.12	2 (7.7)	1623 (6.9)		0.43	7 (5.6)	1618 (6.9)	
No	21 994 (93.1)	0.11	24 (92.3)	21 970 (93.1)		0.55	119 (94.4)	21 875 (93.1)	

The numbers in bold indicate those *P* values < .05.

Abbreviations: GBS, group B *Streptococcus*; IAP, intrapartum antibiotic prophylaxis; NS, beta-lactam nonsusceptible; RBLS, reduced beta-lactam susceptibility.

[a]California started submitting GBS isolates from the surveillance catchment area (3 counties) from July 2014.

[b]Georgia (20 counties), Minnesota (entire state), and Oregon (3 counties) submitted GBS isolates from the surveillance catchment area during the entire surveillance period.

[c]Colorado started submitting GBS isolates from the surveillance catchment area (5 counties) from 2000. During 2005–2010, only isolates from children aged <1 year were sent.

[d]Maryland and New Mexico started submitting GBS isolates from the surveillance catchment area (entire state) from 2004.

[e]New York submitted GBS isolates from the surveillance catchment area (7 counties in 1998, 15 counties from 1999) during the entire surveillance period. From 2006, only isolates from children aged <90 days were sent.

[f]Three hundred ninety-three (1.39%) isolates missing information.

[g]Four (0.01%) missing information.

[h]Two (0.01%) missing information.

[i]Two hundred seventy-one (0.96%) missing information.

[j]Fifty-four (0.19%) missing information.

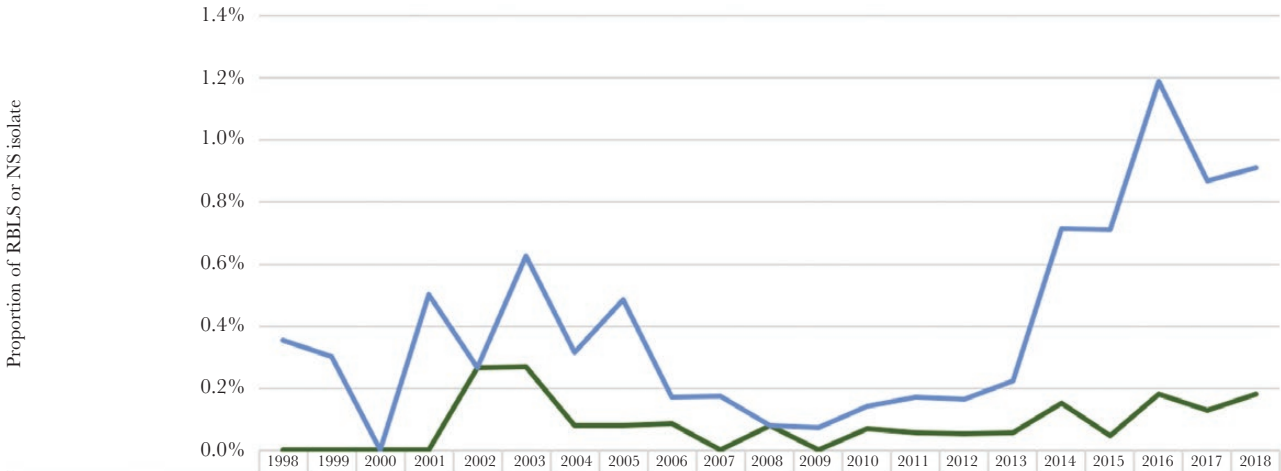
[k]This information started to be collected from year 2009. In infants, syndrome variables were recoded into either bacteremia, meningitis, osteomyelitis, or other, which are mutually exclusive. Two infants who had both meningitis and osteomyelitis were classified as osteomyelitis. These 2 infants did not meet either CLSI or RBLS criteria.

[l]Four hundred five (1.43%) infants missing information.

[m]In adults, syndrome variables were recoded into either bacteremia, cellulitis, osteomyelitis, pneumonia, or other, which are mutually exclusive. Eight hundred seven had both cellulitis and osteomyelitis and were classified as osteomyelitis. Two hundred ninety-four had both pneumonia and cellulitis and were classified as cellulitis. Fifty had both pneumonia and osteomyelitis and were classified as osteomyelitis. Fifteen had pneumonia, osteomyelitis, and cellulitis and were classified as osteomyelitis.

[n]This information started to be collected from year 2000. Eight hundred eighty-seven (3.14%) missing information for adults aged ≥30 years.

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	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018
Total number of isolates	281	331	320	597	753	1116	1261	1237	1158	1142	1224	1350	1411	1733	1824	1776	1962	2109	2188	2302	2194
Number of NS isolates	0	0	0	0	2	3	1	1	1	0	1	0	1	1	1	1	3	1	4	3	4
Number of RBLS isolates	1	1	0	3	2	7	4	6	2	2	1	1	2	3	3	4	14	15	26	20	20
NS (%)	0.0%	0.0%	0.0%	0.0%	0.3%	0.3%	0.1%	0.1%	0.1%	0.0%	0.1%	0.0%	0.1%	0.1%	0.1%	0.1%	0.2%	0.0%	0.2%	0.1%	0.2%
RBLS (%)	0.4%	0.3%	0.0%	0.5%	0.3%	0.6%	0.3%	0.5%	0.2%	0.2%	0.1%	0.1%	0.1%	0.2%	0.2%	0.2%	0.7%	0.7%	1.2%	0.9%	0.9%

Figure 1. Proportion of group B *Streptococcus* isolates with reduced beta-lactam susceptibility[a] and beta-lactam nonsusceptibility[b] by year, ABCs 1998–2018. A, GBS isolates with minimum inhibitory concentration values that exceeded the predefined breakpoints for at least 1 of the 6 beta-lactam antibiotics (ampicillin MIC ≥ 0.5 µg/mL, penicillin MIC ≥ 0.25 µg/mL, cefotaxime MIC ≥ 0.25 µg/mL, ceftizoxime MIC ≥ 1 µg/mL, cefazolin MIC ≥ 1 µg/mL, or cefoxitin MIC ≥ 16 µg/mL). B, Defined as ampicillin MIC > 0.25 µg/mL, penicillin MIC > 0.12 µg/mL, or cefotaxime MIC > 0.5 µg/mL. Abbreviations: ABCs, Active Bacterial Core surveillance; GBS, group B *Streptococcus*; MIC, minimum inhibitory concentration; NS, beta-lactam nonsusceptible; RBLS, reduced beta-lactam susceptibility.

(Supplementary Table 5). During 2015–2018 when all isolates underwent WGS, 66/143 (46.2%) that were PBP2x types ≥6 were non-RBLS; 5/66 (7.6%) had *pbp2x* mutations associated with RBLS (Supplementary Tables 1 and 6; Supplementary Data).

Of the RBLS isolates with information available, diverse PBP2x (Supplementary Figure 2a) and ST (Supplementary Figure 2b) types were observed. Of the 135 isolates with PBP2x information available, there were 68 different PBP2x types, including 48/68 (70.6%) that appeared only once during the study period. Of the 132 isolates with ST information available, ST23 (37/132; 28.0%), ST1 (33/132; 25.0%), and ST19 (11/132; 8.3%) were the most common sequence types. Phylogenetic analysis of *pbp2x* point mutants of the ST1 (Supplementary Figure 3a) and ST23 (Supplementary Figure 3b) lineages showed that specific *pbp2x* mutations have occurred sporadically as multiple independent events within individual strains, rather than significant clonal expansion of a single strain with RBLS.

DISCUSSION

We provide a unique, population-based description of RBLS isolates identified among >28 000 invasive GBS isolates collected over 20 years in the United States. The ABCs platform allowed both clinical and strain characterization of these RBLS cases and harnessed the power of WGS along with phenotypic susceptibility testing to shed light on the emergence of RBLS GBS strains. The proportion of NS isolates remained stable and low. A low but increasing proportion of strains were noted to have first-step *pbp2x* mutations associated with reduced beta-lactam susceptibility. This increase, most marked in recent years, requires further monitoring.

Most RBLS isolates were from older adults, a group that comprises the majority of invasive GBS cases. Mutations conferring RBLS appear to have arisen independently with no notable amplification of a single clone. That said, *pbp2x* mutations associated with RBLS occurred most often within ST1 and 23, which are unrelated but common among US invasive GBS strains [8]. One hypothesis is that individual clonal complexes have underlying genomic traits that predispose for specific mutations within the essential *pbp2x* gene in response to beta-lactam selective pressure. In *Streptococcus pyogenes*, where RBLS and an associated *pbp2x* substitution were recently detected, missense SNPs genomically unlinked to *pbp2x* were found within the *pbp2x* point mutants [18]. Resistance-conferring mutations in PBP genes have been described to increase fitness costs in *S. pneumoniae*, and unlinked alterations within non-PBP genes are likely to alleviate these costs in streptococci [19–21].

Despite the lack of evidence of clonal expansion, we observed an increase in the proportion of RBLS isolates over time, particularly during 2013–2018. We explored several possibilities. First, ceftizoxime, a sensitive marker for RBLS screening [10], was added as part of the routine beta-lactam susceptibility panel in late 2013. However, the increasing trend after 2013

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Table 2. Characteristics of Antibiotic Use for ABCs GBS Cases With NS or RBLS With Information Available (n = 115)

Characteristics	NS Cases (n = 21)[a]	RBLS Cases (n = 115)
Antibiotic treatment for the index GBS infection, N (%)	21 (100)	113 (98.3)
Received antibiotic regimen including beta-lactam antibiotics, N (%)	21 (100)	105 (92.2)
Number of patients who died	0	4 (3.5)
Median duration of antibiotic treatment among those who survived (IQR), d	15 (10–30)[b]	14 (9–18)[c]
Adults aged ≥30 y, N (%)	19 (90.5)	106 (92.2)
Received beta-lactam antibiotics within the previous year, n/N (%)	8/19 (42.1)[d]	37/106 (34.9)[e]
Total number of days of beta-lactam antibiotic treatment in the past year, median (IQR)	25.5 (8–33)[f]	12.5 (2–28.5)[g]
Adults aged ≥30 y with bacteremia, N (%)	2/19 (10.5)	30/106 (28.3)
Received beta-lactam antibiotics within the previous year, n/N (%)	0/2 (0)	13/30 (43.3)
Total number of days of beta-lactam antibiotic treatment in the past year, median (IQR)	NA	5.5 (1–15)[h]
Adults aged ≥30 y with osteomyelitis, N (%)	8/19 (42.1)	24/106 (22.6)
Received beta-lactam antibiotics within the previous year, n/N (%)	5/8 (62.5)[i]	12/24 (50.0)[d]
Total number of days of beta-lactam antibiotic treatment in the past year, median (IQR)	32 (8–33)[j]	21 (18–32)[k]
Adults aged ≥30 y with diabetes, N (%)	16/19 (84.2)	65/106 (61.3)
Received beta-lactam antibiotics within the previous year, n/N (%)	8/16 (50.0)[l]	19/65 (29.2)[m]
Total number of days of beta-lactam antibiotic treatment in the past year, median (IQR)	25.5 (8–33)[f]	18 (6–29)[n]

Abbreviations: GBS, group B *Streptococcus*; IQR, interquartile range; NS, beta-lactam nonsusceptible; RBLS, reduced beta-lactam susceptibility.

[a]The 21 NS cases are also part of the 115 RBLS cases.

[b]One missing information on duration of antibiotic treatment.

[c]Seventeen missing information on duration of antibiotic treatment.

[d]Eight unknown or missing information on history of beta-lactam antibiotic use in the previous year.

[e]Forty-one unknown or missing information on history of beta-lactam antibiotic use in the previous year.

[f]Based on information from 6 patients.

[g]Based on information from 28 patients.

[h]Based on information from 10 patients.

[i]Three unknown or missing information on history of beta-lactam antibiotic use in the previous year.

[j]Based on information from 3 patients.

[k]Based on information from 7 patients.

[l]Seven unknown or missing information on history of beta-lactam antibiotic use in the previous year.

[m]Thirty unknown or missing information on history of beta-lactam antibiotic use in the previous year.

[n]Based on information from 15 patients.

was observed even after excluding isolates determined as RBLS based on ceftizoxime only. Second, the California ABCs site began submitting GBS isolates to the CDC in 2014. In total, isolates from California had the highest proportion of RBLS compared with isolates from other sites. However, the total number of isolates submitted from California was relatively small compared with other sites, and this trend did not change even after excluding California. Third, the primary screening methods to detect RBLS changed in 2015 with the introduction of routine WGS. However, given that RBLS is based on phenotypic criteria, the likelihood of missing isolates with the RBLS phenotype, in theory, should be greater after 2015, when only a subset of isolates were subject to broth microdilution, compared with 1998–2015, when all isolates underwent broth microdilution. Lastly, there may be an increasing proportion of invasive GBS infections among adults who are more likely to have prolonged antibiotic exposure. Over time, cases among adults with underlying conditions such as diabetes increased, and the proportion of adults with syndromes such as osteomyelitis, which accounted for a higher proportion of RBLS cases compared with other syndromes, increased. Osteomyelitis is often challenging to treat and results in prolonged courses of antibiotic treatment, especially in people with diabetes [22]. Although our data were limited to RBLS cases, adults with osteomyelitis tended to be exposed to beta-lactam antibiotics longer compared with those who presented with bacteremia, suggesting that invasive GBS infections in general may be increasing among those who are more likely to have prolonged antibiotic exposure. Regardless, given that the increase in the proportion of RBLS isolates coincided with changes in methodology, additional years of monitoring using the same methodology will be needed to definitively determine whether the proportion of RBLS cases is increasing.

Our study is subject to several limitations. First, the methods used to detect RBLS isolates changed during the study period. To minimize misclassification of RBLS due to the change in methods, we reviewed the 2 RBLS detection methods as described, including application of both methods to ~2000 GBS isolates from 2015 and earlier. We uncovered 2 issues: (1) >10% of RBLS isolates were PBP2x type 1–5 not associated with *pbp2x* mutations conferring RBLS and (2) a small proportion of PBP2x type ≥6 isolates that were phenotypically susceptible to beta-lactam antibiotics contained *pbp2x* mutations associated with RBLS (Supplementary Table 1). *pbp2x* mutations are

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Table 3. Phenotypic Characterization and Diversity of Represented PBP2x Types of Invasive Group B Streptococcus Strains With Reduced Beta-lactam Susceptibility, ABCs 1998–2018 (n = 137)

Number of Antibiotics with RBLS detected	Strain With RBLS to Individual Beta-lactam Antibiotics Determined by MIC Breakpoints						Total Number of RBLS Isolates (%) (n = 137) (	PBP2x 1–5	PBP2x ≥6	PBP2x Types	*pbp2x* Mutation Type[a] (Number of Isolates With the Mutation)
	AMP	PEN	TAX	CZL	ZOX	FOX					
1						Y[c]	35 (25.6)[b]	5	29	1, 103, 104, 105, 106, 107, 110, 122, 135, 26, 5, 58, 80, 85, 90, 92, 95, 99	G415E (13), F486L (2), A471V (1), A514T (1), A616V (1), N684S (1), Q412R (1), R519L (1), S353F (1), T409N (1)
					Y[c]		19 (13.9)	2	17	109, 17, 2, 21, 24, 32, 4, 40, 42, 47, 59, 68, 70, 91, 98	F399V (4), T394A (2), A400E (1), A400G (1), A400T (1), A400V (2), F395C (1), G406D (1), T394I (1), V405A (1), T525I (1)
			Y[c]				2 (1.5)	1	1	1, 33	R433C (1)
	Y[d]						2 (1.5)	1	1	1, 48	A514G+ N575K (1)
2					Y[c]	Y[c]	29 (21.2)[b]	2	26	17, 24, 1, 26, 32, 10, 11, 125, 132, 15, 29, 35, 37, 42, 55, 60, 70, 81, 83, 87	A400V (6), T394I (3), 512-Y-522 insertion (1), A550V (1), Del K358 (1), L534S (1), Q557E (1), V525I (3), A400T (2), G415E (2), A400V+ V525I (1), F399Y (1), S456P (1), T473M (1)
			Y[c]			Y[c]	4 (2.9)	3	1	1, 136, 4	T473M (1)
			Y[c]		Y[c]		2 (1.5)	0	2	120, 121	T555A (1)
				Y[c]	Y[c]		1 (0.7)	0	1	59	G406D (1)
				Y[c]		Y[c]	1 (0.7)	1	0	4	NA
3				Y[c]	Y[c]	Y[c]	6 (4.4)	0	6	10, 17, 100, 35	A400V (3), L534S (2), T573A (1)
			Y[c]		Y[c]	Y[c]	3 (2.2)	0	3	11, 32, 44	A400T+Q412K (1), Q557E (1), T394I (1)
	Y[d]				Y[c]	Y[c]	1 (0.7)	0	1	47	F399V (1)
4			Y[c]	Y[c]	Y[c]	Y[c]	7 (5.1)	0	7	10, 112, 13, 38, 39, 8	L534S (2), T394I (1), T555S (2), G398A (1), T394I+T555S (1), V541F (1)
		Y[d]		Y[c]	Y[c]	Y[c]	3 (2.2)	0	3	102, 59, 98	E257D+T573S+V405A (1), G406D (1), V405A (1),
		Y[d]	Y[c]		Y[c]	Y[c]	2 (1.5)	0	2	11, 94	Q557E (1), L534S (1)
	Y[d]	Y[d]		Y[c]	Y[c]		1 (0.7)	0	1	59	G406D (1)
5		Y[d]	Y[c]	Y[c]	Y[c]	Y[c]	6 (4.4)	0	6	11, 111, 117, 17, 63	Q557E (3), A400V (1)
	Y[d]	Y[d]	Y[c]		Y[c]	Y[c]	3 (2.2)	0	3	39, 59, 99	G406D (1), G398A (1)
	Y[d]	Y[d]		Y[c]	Y[c]	Y[c]	2 (1.5)	0	2	66	G406D (2)
	Y[d]	Y[d]	Y[c]	Y[c]	Y[c]		1 (0.7)	0	1	59	G406D (1)
	Y[d]		Y[c]	Y[c]	Y[c]	Y[c]	1 (0.7)	0	1	10	L534S (1)
6	Y[d]	Y[d]	Y[c]	Y[c]	Y[c]	Y[c]	6 (4.4)	0	6	101, 14, 34, 59, 66, 9	G406D (3), N575D (2), T394I (1)

Abbreviations: AMP, ampicillin; CZL, cefazolin; FOX, cefoxitin; GBS, group B *Streptococcus*; MIC, minimum inhibitory concentration; NS, beta-lactam nonsusceptible; PBP2X, penicillin binding protein 2X; PEN, penicillin; RBLS, reduced beta-lactam susceptibility; TAX, cefotaxime; ZOX, ceftizoxime.

[a]*pbp2x* mutations previously associated with RBLS. See Supplementary table 1.

[b]One isolate missing PBP2x type.

[c]Meets RBLS criteria, defined as ampicillin MIC ≥ 0.5 µg/mL, penicillin MIC ≥ 0.25 µg/mL, cefotaxime MIC ≥ 0.25 µg/mL, ceftizoxime MIC ≥ 1 µg/mL, cefazolin MIC ≥ 1 µg/mL, or cefoxitin MIC ≥ 16 µg/mL.

[d]Meets both NS and RBLS criteria, defined as ampicillin MIC ≥ 0.5 µg/mL, penicillin MIC ≥ 0.25 µg/mL, or cefotaxime MIC > 0.5 µg/mL.

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considered essential for any significantly elevated beta-lactam MICs, and to our knowledge there are no known clinical isolates with elevated beta-lactam MICs that lack *pbp2x* gene alterations; however, mutations at other loci may also contribute to the RBLS phenotype. Additionally, antibiotic information was not available for non-RBLS cases, so we were unable to compare antibiotic treatment history between RBLS and non-RBLS cases. We also lacked population controls to assess whether beta-lactam antibiotic exposure in RBLS cases was higher than that observed in the general public. Lastly, our analyses were restricted to the 8 ABCs sites with isolates available for evaluation, so our results may not be nationally representative.

Early detection of emerging antimicrobial resistance profiles, such as beta-lactam resistance among GBS, is critical for containing resistance before it becomes widespread. Use of CLSI breakpoints suggested a reassuring picture for beta-lactam susceptibility among invasive GBS isolates including beta-lactam antibiotics that are commonly used for the prevention or treatment of invasive GBS cases, including ceftriaxone. Our RBLS criteria revealed the emergence of first-step *pbp2x* mutations that conferred reduced beta-lactam susceptibility among GBS causing invasive disease in the United States, which, although in a small proportion, was associated with a concerning pattern of increase of this phenotype since 2014. With continued exposure to beta-lactam antibiotics, these strains may accumulate additional PBP or non-PBP mutations to achieve higher resistance levels. Given the increasing incidence of invasive GBS disease among older adults [2], subgroups at increased risk for invasive GBS disease, for GBS disease caused by RBLS, and for prior antibiotic exposures, should be monitored closely to detect increases in MIC of RBLS strains and/or increases in the proportion of strains with RBLS. Although we have not found evidence of longstanding NS lineages, we should continue to be vigilant, as such lineages would indicate a level of successful adaptation to beta-lactam selective pressure.

Supplementary Data

Supplementary materials are available at Open Forum Infectious Diseases online. Consisting of data provided by the authors to benefit the reader, the posted materials are not copyedited and are the sole responsibility of the authors, so questions or comments should be addressed to the corresponding author.

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Patient consent. The design of the work has been approved by local ethical committees.

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References

1. Nanduri SA, Petit S, Smelser C, et al. Epidemiology of invasive early-onset and late-onset group B streptococcal disease in the United States, 2006 to 2015: multistate laboratory and population-based surveillance. JAMA Pediatr **2019**; 173:224–33.
2. Francois Watkins LK, McGee L, Schrag SJ, et al. Epidemiology of invasive group B streptococcal infections among nonpregnant adults in the United States, 2008–2016. JAMA Int Med **2019**; 179:479–88.
3. Prevention of group B streptococcal early-onset disease in newborns: ACOG Committee Opinion, Number 782. Obstet Gynecol **2019**; 134:e19–40.
4. Baddour LM, Wilson WR, Bayer AS, et al; American Heart Association Committee on Rheumatic Fever, Endocarditis, and Kawasaki Disease of the Council on Cardiovascular Disease in the Young, Council on Clinical Cardiology, Council on Cardiovascular Surgery and Anesthesia, and Stroke Council. Infective endocarditis in adults: diagnosis, antimicrobial therapy, and management of complications: a scientific statement for healthcare professionals from the American Heart Association. Circulation **2015**; 132:1435–86.
5. Berkowitz K, Regan JA, Greenberg E. Antibiotic resistance patterns of group B streptococci in pregnant women. J Clin Microbiol **1990**; 28:5–7.
6. Dahesh S, Hensler ME, Van Sorge NM, et al. Point mutation in the group B streptococcal pbp2x gene conferring decreased susceptibility to beta-lactam antibiotics. Antimicrob Agents Chemother **2008**; 52:2915–8.
7. Kimura K, Suzuki S, Wachino J, et al. First molecular characterization of group B streptococci with reduced penicillin susceptibility. Antimicrob Agents Chemother **2008**; 52:2890–7.
8. McGee L, Chochua S, Li Z, et al. Multistate, population-based distributions of candidate vaccine targets, clonal complexes, and resistance features of invasive group B streptococci within the US: 2015–2017. Clin Infect Dis. **In press**.
9. Metcalf BJ, Chochua S, Gertz RE Jr, et al. Short-read whole genome sequencing for determination of antimicrobial resistance mechanisms and capsular serotypes of current invasive *Streptococcus agalactiae* recovered in the United States. Clin Microbiol Infect **2017**; 23:574.e7–14.
10. Nagano N, Nagano Y, Toyama M, et al. Penicillin-susceptible group B streptococcal clinical isolates with reduced cephalosporin susceptibility. J Clin Microbiol **2014**; 52:3406–10.
11. Clinical and Laboratory Standards Institute. M100. Performance Standards for Antimicrobial Susceptibility Testing. 29th ed. Wayne, PA: Clinical and Laboratory Standards Institute; **2019**.
12. Langley G, Schaffner W, Farley MM, et al. Twenty years of active bacterial core surveillance. Emerg Infect Dis **2015**; 21:1520–8.
13. Centers for Disease Control and Prevention. Methodology. Available at: https://www.cdc.gov/abcs/methodology/index.html. Accessed 1 June 2020.
14. Centers for Disease Control and Prevention. ABCs isolates sent by surveillance areas, 1997–2018. Available at: https://www.cdc.gov/abcs/downloads/isolates-by-site.pdf. Accessed 3 December 2019.
15. CLSI. Performance Standards for Antimicrobial Susceptibility Testing. *CLSI Supplement M100.* 27th ed. Wayne, PA: Clinical and Laboratory Standards Institute; **2017**.

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16. Slotved HC, Elliott J, Thompson T, Konradsen HB. Latex assay for serotyping of group B *Streptococcus* isolates. J Clin Microbiol **2003**; 41:4445–7.
17. Imperi M, Pataracchia M, Alfarone G, et al. A multiplex PCR assay for the direct identification of the capsular type (Ia to IX) of *Streptococcus agalactiae*. J Microbiol Methods **2010**; 80:212–4.
18. Vannice K, Ricaldi J, Nanduri S, et al. *Streptococcus pyogenes* pbp2x mutation confers reduced susceptibility to β-lactam antibiotics. Clin Infect Dis **2020**; 71:201–4.
19. Rieux V, Carbon C, Azoulay-Dupuis E. Complex relationship between acquisition of beta-lactam resistance and loss of virulence in *Streptococcus pneumoniae*. J Infect Dis **2001**; 184:66–72.
20. Sauerbier J, Maurer P, Rieger M, Hakenbeck R. *Streptococcus pneumoniae* R6 interspecies transformation: genetic analysis of penicillin resistance determinants and genome-wide recombination events. Mol Microbiol **2012**; 86:692–706.
21. Trzcinski K, Thompson CM, Gilbey AM, et al. Incremental increase in fitness cost with increased beta-lactam resistance in pneumococci evaluated by competition in an infant rat nasal colonization model. J Infect Dis **2006**; 193:1296–303.
22. Lipsky BA, Berendt AR, Cornia PB, et al; Infectious Diseases Society of America. Executive summary: 2012 Infectious Diseases Society of America clinical practice guideline for the diagnosis and treatment of diabetic foot infections. Clin Infect Dis **2012**; 54:1679–84.

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